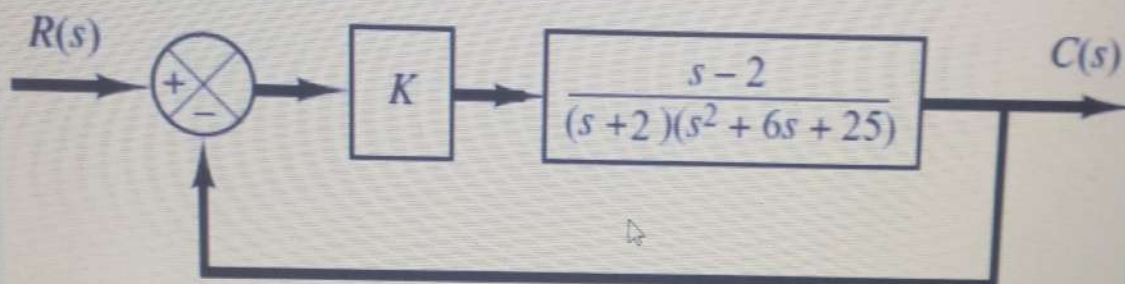


2022-3-ELE0334-CONTROL SYSTEMS-1

es / 2023/2022-3-ELE0334-CONTROL SYSTEMS-1 / General / Quiz 2

Consider the closed-loop system shown in Figure below. Determine the range of K for stability. Assume



Answer

$$-41 < K < 25$$

Awad

Question 1) choose the correct answer

1) Asymptotes of root locus intersect the real axis at for the following function is

$$G(s) = 1 / [s(s+2)((s+2)^2 + 4)]$$

- a) -2.55 b) -1.44 c) -2 d) -1

2) Consider the system represented by the equation given below. $200s^2 / [(s+3)(s+6)(s+10)]$ and be the total phase value at

- a) 90 b) -90 c) 180 d) -180

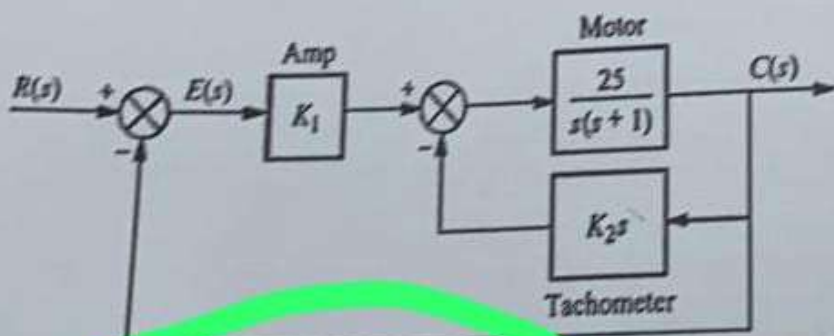
3) Find the centroid of the root locus for the system having $G(s)H(s) = K / (s+4)(s^2+4s+5)$

- a) -2.66 b) -2 c) -8 d) -1.66

4) If a feedback control system has its open loop transfer function $G(s)H(s) = 1 / (s(s+2)((s+2)^2 + 4))$ the root locus plot which intersects the imaginary axis

- a) +j35.4 b) +j13.3 c) +j1.63 d) +j18.25

5) consider the block diagram shown in the next figure. Determine appropriate value for K_1 and K_2 to achieve overshoot 25% and settling time of 5 second for unit step input.



- a) $k_1 = 0.14$ and $k_2 = 0.024$ b) $k_1 = 0.27$ and $k_2 = 0.3$
c) $k_1 = 0.17$ and $k_2 = 0.024$ d) $k_1 = 0.27$ and $k_2 = 0.34$

6) The approximate settling time of the closed loop system is $G(s) = 8 / (s^2 + 4s + 8)$

- a) 1.75s b) 1.57s c) 1.6s d) 1.65s

7) If the system is specified by open loop transfer function $G(s)H(s) = k / s(s+3)(s+2)$, how many poles will proceed to end at infinity?

- a) 1 b) 0 c) 2 d) 3

8) Which condition is used to verify the existence of a particular point on the root locus?

- a) Frequency b) Angle c) Amplitude d) Magnitude

9) The approximate peak percentage overshoot of the closed loop system is $G(s) = 9 / (s^2 + 4s + 8)$

- a) 6% b) 9% c) 6% d) 0.9%

10) if the pole location for the second order system move vertically then the step responses will have

a) same overshoot b) Same damping constant c) same frequency d) nothing change

11) Decreasing the undamped natural frequency will generally increase the rise time of the step response.

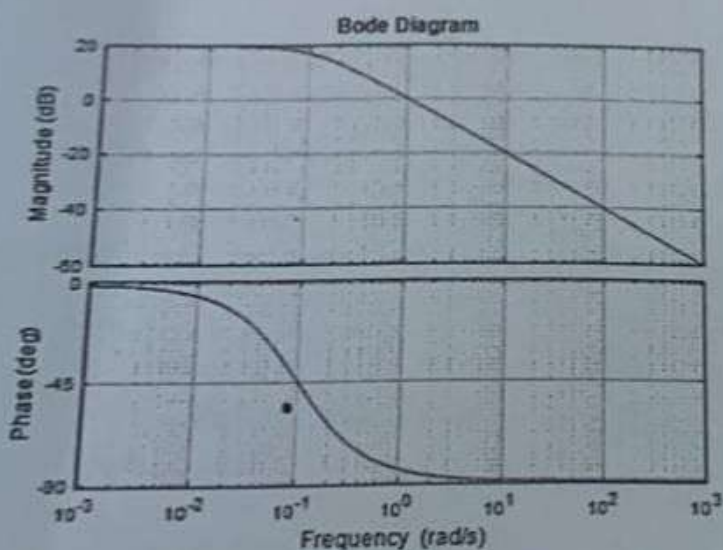
a) true b) false

12) a) Applying impulse response for the first order system below then the DC gain is _____

$$G(s) = 2/(s+0.4)$$

- a) 5 b) 2.5 c) 2 d) 0.4

13) which of the following function represent the figure below



a)

$$L(s) = 0.01 \cdot \frac{(s+10)^2}{s+0.1}$$

b)

$$L(s) = \frac{10}{s+0.1}$$

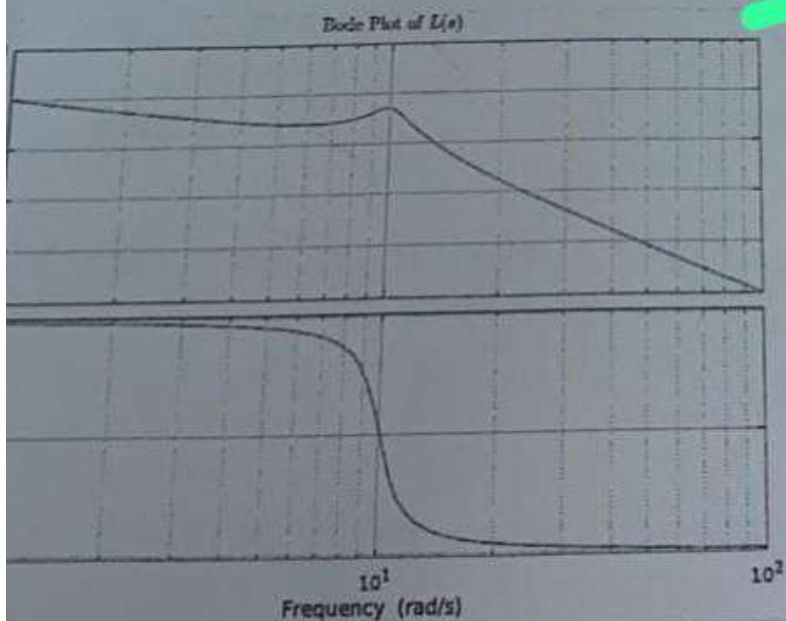
c)

$$L(s) = \frac{10}{0.1-s}$$

d)

$$L(s) = \frac{s+100}{(s+0.1)(s-100)}$$

Consider the following Bode plot for a loop transfer function $L(s)$:



a) $L(s)$ has 2 poles and 0 zeros.

b) $L(s)$ has 4 poles and 2 zeros.

c) $L(s)$ has 3 poles and 2 zeros.

d) $L(s)$ has 3 poles and 0 zeros.

The unit ramp steady state error of a standard first order system is

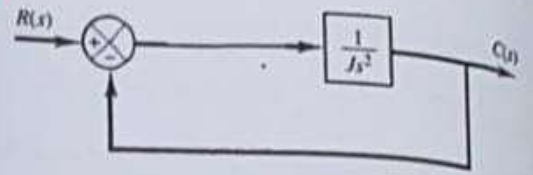
A] 0

B] 3T

C] T

D] 4T

16. The suitable controller for the shown system is
 A) I B) P C) D D) PD



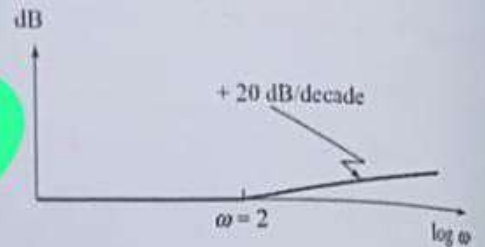
17. The shown dB plot corresponds to:

(A) $\frac{1}{2s+1}$

(B) $2s+1$

(C) $\frac{1}{\frac{1}{2}s+1}$

(D) $\frac{1}{2}s+1$



18. The open loop transfer function of unity feedback control system is $G(s) = K/(s(s+1))$. If gain K is increased to infinity, then the damping ratio will tend to become

A. infinite

B. 0.707

C. unity

D. zero

19. Consider the system represented by the transfer function $200/(s^2 + 2s + 1)$. The total phase angle at $\omega=0$

A. 0

B. -90

C. -180

D. -270

20. The output of first order system given by $C(s)/R(s) = 2/(s+4)$, the steady state error is

A. 0

B. 0.5

C. 1

D. undefined

21. The following controller is seldom (نادراً) used alone in practice:

(A) Proportional

(B) Derivative

(C) Integral

(D) Proportional-Integral-Derivative

22. The closed loop system given by the characteristic equation $3s^3 + 2s^2 + bs + a = 0$ is stable if

A. $a>0; b>0$

B. $a>0; b>0; b<1.5a$

C. $a>0; b>0; b>1.5a$

D. $a>0; b<1.5a$

23. A unity feedback system has open-loop transfer function $G(s) = 2s/[s(s+6)]$. The peak overshoot to step input is:

A] 15%

B] 20%

C] 5%

D] 10%

24. The system given by $GH(s) = \frac{K}{s(s+1)(s+2)}$ has crossing point(s) on the imaginary axis at

A) 0, -1 B) $\pm j\sqrt{2}$ C) $-j\sqrt{2}$ D) $j\sqrt{2}$

25. If the characteristic polynomial of a closed loop system is $s^2 - 2s + 1 = 0$, then the system is

A: critically damped

B: underdamped

C: undamped

E: overdamped

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2) Consider the system represented by the equation given below. What is the total phase value at $\omega = 200$ rad/s?

$$200s^2 / [(s+3)(s+6)(s+10)]$$

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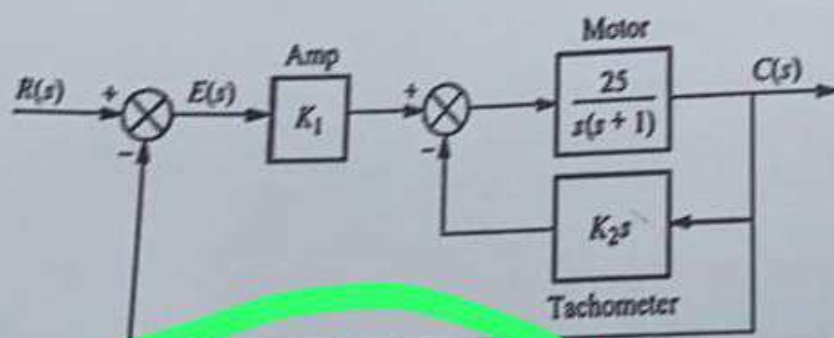
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